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Origin of Layered Herringbone Packing and Polymorphism in Polyacenes: A Quantum Chemical Optimization Approach

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ABSTRACT

The complex molecular packing motifs and polymorphism of polyacenes have long posed significant challenges, despite the apparent simplicity of their individual molecular structures. In this study, we investigate the crystal structures of a series of polyacenes using a stepwise quantum chemical optimization approach, employing dispersion-corrected density functional theory (DFT) to elucidate the origins of their intermolecular arrangements. We begin by modeling a molecular pair in which the short axes are arbitrarily positioned and oriented, while the long axes are kept parallel and unshifted. By imposing glide symmetry, we establish a fundamental repeating unit that enables the construction and optimization of monomolecular layers exhibiting the most stable herringbone arrangement—termed the “regular” *R* form. Subsequent introduction of long-axis inclination yields two locally stable configurations: one preserving glide symmetry (*G* form) and one that does not (*N* form). These optimized structures correspond well with experimental observations: the *R* form matches the pentacene thin-film phase (Type I), the *G* form aligns with naphthalene and anthracene (Type III), and the *N* form reflects the packing of other polyacenes (Types II and IV). Further refinements, including subtle long-axis twisting and interlayer displacements, successfully reproduce all four experimentally observed polymorphs, accurately capturing their structural details. The energetic stability of these phases also aligns with the experimentally observed polymorphic tendencies when interactions with far distant molecules are taken into account. Notably, this approach—combined with potential map analysis—reveals the stabilizing mechanisms specific to each polymorph. These insights further clarify key aspects of charge transport in the crystals. Overall, this work establishes a robust first-principles framework for understanding molecular alignment in polyacene crystals and provides a predictive tool for designing and controlling functional molecular materials.

I. INTRODUCTION

Polyacenes, a class of linearly fused aromatic hydrocarbons, represent archetypal extended π -conjugated systems with significant applications in organic semiconductors (OSCs). This series, ranging from naphthalene ($n = 2$) to hexacene ($n = 6$), has attracted considerable attention due to its exceptional optoelectronic properties in the crystalline phase.¹ Among them, pentacene ($n = 5$) is particularly notable for its high performance in organic field-effect transistors (OFETs), especially when deposited as uniaxially oriented polycrystalline thin films via vacuum deposition techniques.^{2,3} These advances have played a key role in the development of the interdisciplinary field of organic electronics.^{4,5} A defining feature of pentacene and other recently developed OSCs is their pronounced layered crystallinity, which is critical for enabling efficient two-dimensional charge transport and for fabricating high-performance OFET channel layers.^{6–8}

A common packing motif in polyacenes is the layered herringbone (LHB) arrangement, where neighboring planar, rodlike molecules engage in *T*-shaped (core-to-edge) and slipped parallel (core-to-core) contacts.^{9,10} While the LHB motif is commonly conserved across the polyacene series, subtle variations in intermolecular arrangement—and the occurrence of polymorphs^{11–16} in compounds like tetracene and pentacene—reveal an underlying structural complexity that remains poorly understood. Experimentally, polyacene crystal structures can be classified into four distinct types (Fig. 1): (1) a simple LHB arrangement with molecular long axes nearly perpendicular to the layer plane, as observed in the pentacene thin-film phase (Type I);¹⁷ (2) an LHB arrangement with inclined molecular long axes, typical of single-crystal pentacene and hexacene (Type II);^{18,19} (3) a twisted long-axis orientation between adjacent molecules, while retaining glide symmetry, as seen in naphthalene and anthracene (Type III);^{20,21} and (4) an intermediate phase between Type II and III, found in tetracene and polymorphic pentacene (Type IV).^{18,22} These distinct packing motifs form the structural foundation of high-performance OSCs, where even subtle differences in molecular arrangement can significantly impact thin-film morphology and device performance.^{23–30} However, controlling these packing motifs remains a major challenge in crystal engineering,^{31,32} partly because the mechanisms governing the stability of LHB packings and the origins of their polymorphism are still not understood.

Recent advances in quantum chemical calculations have provided powerful tools for probing the intermolecular forces—both attractive and repulsive—that govern molecular crystal formation. While the coupled-cluster singles and doubles model with perturbative triples [CCSD(T)] and a sufficiently large basis set yields highly accurate intermolecular interaction energies,³³ its computational cost remains prohibitive for large molecular systems such as polyacenes. In contrast, density functional theory (DFT), particularly when combined with appropriate functionals and dispersion corrections, has proven to be a practical and effective alternative for analyzing such systems.^{34–39} Nevertheless, a fundamental limitation of quantum

chemical methods is their limited ability to explain why certain packing motifs are energetically favored over others.

This limitation is closely related to the broader challenge of *ab initio* crystal structure prediction—a central pursuit in computational chemistry.^{40–42} Although notable progress has been made, especially in predicting polymorphs of pharmaceutical compounds,⁴³ predicting the crystal structures of complex organic molecules remains a formidable task. Conventional approaches typically involve generating a large number of candidate structures and evaluating their lattice energies to identify the most stable configurations.^{44–50} While empirical force fields have traditionally been used for these calculations, recent advances in dispersion-corrected DFT (DFT-D) methods have significantly improved prediction accuracy.^{51–53} Even so, the high dimensionality of molecular positional and orientational variables often leads to numerous closely related local minima,^{54–62} complicating the accurate prediction of crystal structures even for polyacenes. Moreover, current methods generally lack the ability to explain the underlying origins of crystal stability, an essential aspect for rational crystal engineering.^{31,32} Standard tools such as energy landscape diagrams often fall short in addressing this gap, underscoring the need for a more fundamental, mechanistic understanding.

In this study, we aim to elucidate the stabilization mechanisms underlying polyacene crystal structures, using these systems as models with a particular focus on how packing variations depend on the number of fused rings. To this end, we employed a stepwise optimization approach^{63–65} to model polyacene crystal structures, assuming that these molecules adopt layered arrangements with relatively high symmetry. This assumption allowed us to systematically reduce the complexity of intermolecular configuration variables and optimize them sequentially in order of energetic importance. Through potential map analyses of intermolecular contacts and interaction energies obtained via DFT-D calculations, we successfully reproduced all four experimentally observed polyacene crystal structures as energy-minimized configurations, accurately capturing their structural details. These analyses offer detailed insights into the factors that stabilize each crystal form and provide a deeper understanding of the fundamental principles governing packing motifs and polymorphism in polyacenes.

II. RESULTS

2.1. Optimization of Regular Intralayer Molecular Arrangements.

We began by investigating the simplest molecular packing arrangement—a layered structure in which the long axes of all the polyacene molecules are parallel to each other and perpendicular to the layer surface. Each monomolecular layer was modeled using a basic unit consisting of two adjacent polyacene molecules, aligned with their long axes parallel and unslipped relative to each other, as illustrated in Fig. 2a. Assuming glide symmetry along the long axis, this molecular pair

served as the repeating unit, allowing the entire monomolecular layer to be constructed through a combination of glide reflection and translation operations.^{65,66} The resulting arrangement is described by three independent parameters: the angle (α) between the molecular plane and the glide plane, as well as two orthogonal lattice constants, a and b (Fig. 2a). This configuration forms T -shaped contacts, with an interplanar dihedral angle of 2α , resulting in a two-dimensional arrangement composed of a single crystallographically independent molecule under glide symmetry.

To optimize the intralayer packing arrangement, we systematically explored parameter combinations that minimize the total intermolecular interaction energy (or maximize the stability) of the intralayer structure. First, we performed a geometric search, using van der Waals (vdW) contacts between adjacent molecules to minimize the unit cell size,⁶⁷ as shown in Fig. S1. This crude estimation is practical and reasonable, given that intermolecular interactions between neutral polyacene molecules are primarily governed by the balance between attractive dispersion forces⁶⁸ and repulsive exchange interactions, as discussed later. For each α , we varied a and b to identify candidate structures, which were then used as initial geometries for further optimization via DFT-D calculations.⁶⁹

To compute the total interaction energy, we considered only the eight nearest-neighbor interactions within the intralayer structure—four T -shaped and four slipped-parallel neighbors—as depicted in Fig. S1d. The sum of these interactions is denoted as $E_{\text{intra}(8)}$, which was used to identify stable and metastable structures by optimizing a and b at each α (Fig. S2). (The impact of including intermolecular interactions with more distant molecules will be discussed in Section 2.6.)

Figure 2b shows $E_{\text{intra}(8)}$ for the optimized stable and metastable structures of pentacene at each α , with each point representing a distinct packing motif characterized by specific intermolecular contacts (Fig. 2c). Similar trends were observed across the polyacene series (Fig. S3a). Due to the mirror symmetry of polyacene molecules, these plots are symmetric around $\alpha = 45^\circ$, where the arrangements involve two perpendicular glide planes. The lowest $E_{\text{intra}(8)}$ values were found near $\alpha \approx 25^\circ$ and 65° , corresponding to two equivalent herringbone (HB) structures (Figs. 2c–1 and –5). Notably, the optimized herringbone angle (the angle between the planes of two polyacene molecules), $2\alpha \approx 51^\circ$, closely aligns with the experimentally observed herringbone angle ($\theta_{\text{HB}} \approx 53^\circ$). This stable packing, referred to as the regular (R)-form HB layer, closely resembles the intralayer packing of Type-I crystals (Fig. 1). The R -form serves as a fundamental structural model for further exploration of polyacene crystal structures.

Next, we examined the metastable packing motifs identified during optimization, focusing on alternative arrangements such as π -stacked (PS) and crosshatch (CH) structures. The PS motifs (Figs. 2c–2 and –6) exhibit significantly higher $E_{\text{intra}(8)}$ values than the HB motifs, despite their

compact molecular packing. In contrast, the CH structure, observed at $40^\circ < \alpha < 50^\circ$ (green points in Fig. 2b), lacks slipped-parallel contacts and features larger intermolecular spacings, leading to an increased lattice volume (Fig. 2c–4).

To better understand the relative stability of HB and PS packings, we analyzed the intermolecular interaction energies of six adjacent molecules within the same layer at $\alpha = 25^\circ$ using a radar chart (Fig. 3a). (The interaction with second-nearest slipped-parallel contact is much weaker than these primary interactions in these packings. Hereafter, we denote the sum of primary interactions with the six nearest-neighbor molecules as $E_{\text{intra}(6)}$.) The interactions at the *T*-shaped contact in the HB packing are slightly weaker than those at the slipped-parallel contact in the PS packing. Nonetheless, $E_{\text{intra}(6)}$ is lower in the HB packing than in the PS packing, due to the greater number of *T*-shaped contacts.

Symmetry-adapted intermolecular perturbation theory calculations based on DFT wavefunctions (SAPT(DFT))^{70,71} were employed to decompose the contributions of electrostatic, induction, dispersion, and exchange-repulsion forces to the total intermolecular interactions (Fig. 3b). SAPT(DFT) can calculate these contributions individually. It was demonstrated that the total interaction energy, obtained as the sum of each contribution, is close to that of highly accurate CCSD(T) calculations. It is found in Fig. 3b that the total interaction energy by DFT-D calculations is close to that of SAPT(DFT). The results of energy decomposition suggest that both dispersion forces and electrostatic forces are important for the stability of HB packing, while electrostatic contributions play a relatively minor role. The equilibrium intermolecular distance and total interaction energy are ultimately governed by the balance between attractive forces and exchange repulsion (Fig. 3c).

Figure 4a illustrates how the intermolecular *T*-shaped contacts vary with α in the metastable HB packing (also see Fig. S4). At $\alpha = 25^\circ$, hydrogen atoms at the side edge of one molecule fit precisely into the central hollow of the benzene ring of an adjacent molecule, maximizing carbon-carbon dispersion interactions while minimizing exchange repulsion. This type of interaction is often referred to as CH/ π interaction.^{72,73} However, it is evident that the stability of HB packing is primarily driven by enhanced dispersion forces at the *T*-shaped contacts, particularly at $\alpha = 25^\circ$, where exchange repulsion is minimized. This configuration results in the most stable HB packing and minimizes the lattice volume (Fig. 4b).

Finally, Fig. 4c illustrates how $E_{\text{intra}(8)}/n$ (where n is the number of fused rings) varies across the HB, PS, and CH packings. In the CH packing, $E_{\text{intra}(8)}/n$ is nearly constant, whereas in the HB and PS packings, the magnitude of $E_{\text{intra}(8)}/n$ increases with n . These results likely reflect the distinct nature of intermolecular interaction in polyacenes. In the CH packing, interactions are primarily localized at the *T*-shaped contact between ring planes and ring edges, leading to a nearly linear dependence of $E_{\text{intra}(8)}$ on n . In contrast, the HB and PS packings involve interactions

between ring planes. This feature should result in the slight n -dependence of $E_{\text{intra}(8)}/n$, where the detailed electronic and molecular structures are not obtained as a simple linkage between rings but reflect the characteristics of the whole molecule.

2.2. Optimization by Uniform Long-Axis Inclination.

We investigated the effect of long-axis inclination—a structural feature observed in actual crystals—on molecular packing within polyacene layers. A uniform long-axis inclination of the R -form at $\alpha = 25^\circ$ is represented by the simultaneous sliding of molecules along their long axes while maintaining a flat layer surface (Fig. 5a). Importantly, the intermolecular spacing between molecular planes remains constant during initial optimization. To evaluate the $E_{\text{intra}(6)}$ of the monomolecular layer under these shifts, we performed intermolecular interaction energy calculations, and represent the calculated results using two variables: the inclination angle of the molecular long axis from the normal to the layer surface (θ_{incl}) and the inclination direction within the layer (φ_{incl}). Notably, glide symmetry is preserved when $\varphi_{\text{incl}} = 0^\circ$ or 90° , where inclinations are parallel to the glide plane, but is lost at intermediate φ_{incl} values.

The $E_{\text{intra}(6)}$ landscape for the pentacene monomolecular layer, plotted as a function of θ_{incl} $\cos(\varphi_{\text{incl}})$ and θ_{incl} $\sin(\varphi_{\text{incl}})$, is shown in Fig. 5b. The potential map reveals multiple local minima. The global minimum occurs at $\theta_{\text{incl}} = \varphi_{\text{incl}} = 0^\circ$, corresponding to the original R form. A second local minimum appears at $\theta_{\text{incl}} \approx 25.4^\circ$ (denoted θ_{incl}^N) with four equivalent minima at $\varphi_{\text{incl}} = \pm\varphi_{\text{incl}}^N$ and $\pm(180^\circ - \varphi_{\text{incl}}^N)$, where $\varphi_{\text{incl}}^N \approx 47.6^\circ$, corresponding to the N form, which lacks glide symmetry. A third local minimum appears at $\theta_{\text{incl}} \approx 32.6^\circ$ (denoted θ_{incl}^G) at $\varphi_{\text{incl}} = 0^\circ$ and 180° , corresponding to the glide-symmetric G form. Remarkably, these minima appear consistently across all polyacenes, with only slight variations in θ_{incl} and φ_{incl} values (Fig. S5). Schematic representations of the inclined N and G forms are shown in Fig. 5c. Our findings suggest that the N form closely replicates the packing motifs observed in Type II and IV crystals, while the G form corresponds to the packing found in Type III crystals.

Figure S6a illustrates the relative long axis shifts between six neighboring molecules in the locally stable N and G forms of pentacene. In the N form, the long-axis shift (Δz_T) between adjacent T -shaped contacts is given by

$$\Delta z_T = \frac{ab}{\sqrt{a^2+b^2}} \tan \theta_{\text{incl}}^N \approx 2.4 \text{ \AA},$$

which is approximately the length of one acene unit. In the G form, the shift is

$$\Delta z_T = \frac{b}{2} \tan \theta_{\text{incl}}^G \approx 2.4 \text{ \AA}.$$

These alignments indicate that the stability of these forms is inherently linked to the molecular structure of polyacenes.⁶³

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6 Additionally, Fig. S6b shows variations in interaction energies for *T*-shaped and slipped-
7 parallel contact molecule pairs with long-axis shifts. The interaction energy for the *T*-shaped
8 contact pair increases significantly with a half-ring-length slide, likely due to increased exchange
9 repulsion as hydrogen atoms approach the C–C bonds of adjacent molecules. However, a full-
10 ring-length slide restores a relatively stable configuration, aligning hydrogen atoms with the
11 central hollow of the benzene ring in an adjacent molecule. This configuration maximizes C–C
12 dispersion interactions while minimizing exchange repulsion. In contrast, the interaction energy
13 at slipped-parallel contacts remains largely unchanged by these shifts, becoming marginally more
14 stable with a half-unit slide, possibly due to reduced repulsion or weak electrostatic interactions.

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19 The local stability of the *N* and *G* forms arises from these energy variations. In the *N* form,
20 one-ring-length misalignment occurs at *T*-shaped contacts between two adjacent molecules,
21 whereas in the *G* form, it is present in all four adjacent molecules. This results in the observed
22 stability hierarchy: $E_{\text{intra}(6)}$ for the *R* form is lowest, followed by the *N* form, with the *G* form being
23 the least stable. The stability of the *N* form depends on the balance between *T*-shaped and slipped-
24 parallel contacts. Further optimization of the *N* and *G* forms, treating *a* and *b* as additional
25 variables, confirmed that the *G* form remains locally stable across all polyacenes. However, for
26 shorter polyacenes ($n = 2$ and 3), the *N* form ceases to be a local minimum due to insufficient
27 long-axis shifts at *T*-shaped contacts, which increases exchange repulsion. This instability
28 explains why Type III crystals, associated with the *G* form, are predominantly observed in shorter
29 polyacenes like naphthalene and anthracene. As shown in Fig. 5d, the stable θ_{incl} increases with
30 molecular length for longer polyacenes ($n \geq 4$). These subtle variations create competition
31 between Type II and IV crystals, as will be discussed in Sections 2.3 and 2.5.

32 33 34 35 36 37 38 39 40 41 **2.3. Optimization of Interlayer Stacking Arrangements.**

42 As a third step in our structural optimizations, we investigated interlayer stacking
43 arrangements between monomolecular layers derived from previous optimizations. This stepwise
44 approach is quite effective, since interlayer interactions are much weaker than intralayer
45 interactions. (An analysis of this feature will be presented in Section 2.6.) Notably, further re-
46 optimization of intralayer molecular arrangements becomes useful after considering interlayer
47 interactions, as discussed in Sections 2.4 and 2.5.

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51 To determine the optimal interlayer stacking, we systematically searched for the lateral
52 displacement coordinates (x, y) between parallel layers (Fig. 6a). First, we calculated the interlayer
53 distance z at vdW contacts for each coordinate, since interlayer distance is directly linked to
54 crystal stability, with dispersion forces playing a dominant role. The upper panels of Fig. 6b–d
55 present the calculated unit cell volumes $V(x, y)$ for the *R*, *G*, and *N* forms, respectively, providing
56 an overview of stable interlayer arrangements, which are highly sensitive to the mutual
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engagement of surface geometries between layers.

For the *R*-form, the unit cell volume $V(x, y)$ exhibits symmetric behavior, satisfying $V(x, y) = V(\pm x, \pm y)$. The optimal interlayer engagement (*i.e.*, the configuration with minimal unit cell volume) is found at $(x, y) = (\pm 1.8 \text{ \AA}, -0.6 \text{ \AA} \sim +0.6 \text{ \AA})$, highlighted by the blue-banded region in the upper panel of Fig. 6b. Next, we further optimized the interlayer distance and lateral coordinates to minimize the sum of intermolecular interactions. To compute the total interlayer interaction energy, we considered the seven nearest-neighbor interactions across the layers; one interlayer nearest neighbor and its six intralayer nearest neighbors, as depicted in Fig. S7. The sum of these interactions, denoted as $E_{\text{inter}(7)}$, was used to identify stable and metastable interlayer arrangements. $E_{\text{inter}(7)}$ exhibits a broad minimum at $(\pm 1.8 \text{ \AA}, 0)$ (lower panel of Fig. 6b), resulting in a crystal structure closely resembling the Type-I structure. However, a key difference in crystal symmetry remains compared to the actual thin-film phase of pentacene, which will be discussed in Section 2.7.

For the *G*-form of naphthalene (Fig. 6c), the calculated unit cell volume $V(x, y)$ exhibits inversion symmetry with respect to $y = 0$, following $V(x, y) = V(x, \pm y)$, due to the glide symmetry of the *G*-form monomolecular layer, which has a glide plane parallel to the x axis (see Fig. 5c). The volume reaches a minimum at $(x, y) = (1.9 \text{ \AA}, 0 \text{ \AA})$, where the surface geometries of the layers align optimally. Note that the lateral coordinate of the central molecule in the stacked layer was set as the origin, ensuring that central molecules coincide between layers along molecular long-axis directions. This coordinate system allows for a clear visualization of interlayer relationships (see Figs. 1 and 6e). Further optimization using DFT-D yielded the final structure, shown in the lower panel of Fig. 6c. (This structure corresponds to the “twisted *G*” form, detailed in Section 2.4.)

For the *N* form of pentacene (Fig. 6d), the calculated unit cell volume $V(x, y)$ does not exhibit inversion symmetry. The plot reveals two distinct local minima at $(x, y) = (0.0 \text{ \AA}, 1.4 \text{ \AA})$ and $(1.4 \text{ \AA}, -0.7 \text{ \AA})$, labelled N_1 and N_2 , respectively. Here, the molecular edges engage smoothly with the concave surface of the opposing layer, forming well-aligned interlayer stacks along local minimum valleys (light blue regions). These local minima correspond closely to the Type-II and Type-IV structures of polyacenes, respectively, as shown in the comparison between Figs. 1 and 6e. In pentacene ($\theta_{\text{incl}} = 22.1^\circ$), $E_{\text{inter}(7)}$ is slightly lower for the N_1 stacking (Type II) than for N_2 stacking (Type IV), although the energy difference is small.

We observed that the relative stability of the N_1 and N_2 stacking arrangements varies with the ring number n due to changes in the optimized intralayer inclination angle θ_{incl} (Fig. 5d). The lower panels of Fig. 6d show the optimization results for interlayer stacking in the *N* form of tetracene, calculated at two inclination angles: $\theta_{\text{incl}} = 25.4^\circ$ and $\theta_{\text{incl}} = 20.0^\circ$, corresponding to the optimized values for hexacene and tetracene, respectively (Fig. 5d). The N_1 stacking arrangement

(Type II) becomes relatively more stable at $\theta_{\text{incl}} = 25.4^\circ$, while the N_2 arrangement (Type IV) is favored at $\theta_{\text{incl}} = 20.0^\circ$. These results provide a plausible explanation for the ring-number dependence of polymorph formation in the experimental observations: Type-II crystals (N_1 stacking) are more common in longer polyacenes ($n = 5$ and 6), while Type-IV crystals (N_2 stacking) tend to form in medium-length polyacenes ($n = 4$ and 5).

2.4. Long-axis Twist Under Glide Symmetry.

The Type-III structure, observed in shorter polyacenes such as naphthalene and anthracene, is characterized by a noticeable non-parallel alignment of molecular long axes at T -shaped contacts. Despite this inclination, these crystals retain glide symmetry, distinguishing them from longer polyacenes. As described in Section 2.2, the G -form structure maintains glide symmetry through a uniform long-axis inclination at $\varphi_{\text{incl}} = 0^\circ$. In this section, we investigate the Type-III structure using G -form monomolecular layers as the starting point.

To model the Type-III structure, we introduced a twist shift, in which the molecular plane is rotated by an angle θ_{twist} around an axis perpendicular to the plane (Fig. 7a), combined with a sliding shift along the long axis of the original G form. Glide symmetry is preserved by rotating the adjacent molecule at the T -shaped contact in the opposite direction, as illustrated in Fig. 7b. The intralayer arrangements were optimized by minimizing $E_{\text{intra}(6)}$, obtained from DFT calculations, for each θ_{twist} with a , b , and Δz_T as variable parameters (instead of θ_{incl} for clarity). As shown in Fig. 7c, the optimal Δz_T for the G form decreases as θ_{twist} increases. The lowest $E_{\text{intra}(6)}$ values were obtained at $\theta_{\text{twist}} \approx 14^\circ$ for naphthalene ($n = 2$) and $\theta_{\text{twist}} \approx 3^\circ$ for anthracene ($n = 3$) (lower panel of Fig. 7c). Notably, in longer polyacenes ($n = 4, 5, 6$), no energy decrease was observed with this operation, suggesting that non-parallel long-axis inclinations stabilize the structure only when the molecular length is sufficiently short.

Next, we examined the role of interlayer interactions in these optimized monomolecular layers. We evaluated the total energy of intralayer and interlayer interactions with adjacent molecules, denoted as,

$$E_{\text{int(near)}} = E_{\text{intra}(6)} + 2 \times E_{\text{inter}(7)}.$$

We minimized $E_{\text{int(near)}}$ for each θ_{twist} with a , b , Δz_T , x , y , and z as variable parameters using DFT calculations. The results (Fig. 7d) show that interlayer stabilization increases significantly with θ_{twist} . Importantly, the obtained optimal twist angles, $\theta_{\text{twist}} = 13^\circ$ for naphthalene and $\theta_{\text{twist}} = 9^\circ$ for anthracene, closely match those observed in experimental crystal structures. Additionally, we confirmed that the parameter α had little impact on stabilization, indicating that the dihedral angle between molecular planes, or the herringbone angle $\theta_{\text{HB}} = 2\alpha$, remains nearly unchanged despite the introduction of non-parallel long-axis inclinations. This observation is consistent with actual crystals.

The decrease in $E_{\text{int(near)}}$ due to twisting arises from geometric modifications at T -shaped contacts. Notably, the critical interaction between the benzene ring and a hydrogen atom—essential for G form stability—remains intact even after the twist shift. Furthermore, an additional interaction forms between another hydrogen atom and the benzene ring within the same molecular pair (Fig. 7b, right panel). The twist shift also alters the T -shaped arrangement, widening (and alternately narrowing) the molecular gap at the layer surface, allowing the long-axis edge of the molecule to fit the gap more effectively (Fig. 7e). As a result, the greatest stabilization via θ_{twist} occurs at the smallest lattice volumes, aligning with the concept of close packing. However, in longer polyacenes, this modification is hindered, as destabilizing effects from the non-parallel configuration outweigh the stabilization gain.

2.5. Further Optimization with Non-Uniform Long-Axis Inclination.

The Type-IV structure, observed in the bulk phase of tetracene^{74,75} and in a pentacene polymorph formed by heating,⁷⁶ is characterized by N_2 -type interlayer stacking. In actual Type-IV crystals, slight non-uniform (or non-parallel) long-axis inclinations are present, though these effects were not previously considered in the optimization process. To further investigate the stability of Type-IV structures, we introduced independent long-axis inclinations to the N_2 -type arrangement while preserving inversion symmetry. This approach allowed us to assess the relative stability between Type-II and Type-IV structures in polyacenes.

In this model, only the central molecule in the unit cell is inclined by an angle θ'_{incl} and direction φ'_{incl} , while the adjacent molecule remains fixed (Fig. 8a). Figure 8b shows the $E_{\text{intra(6)}}$ values for the N -type monomolecular layer of tetracene, plotted as functions of θ'_{incl} and φ'_{incl} , with the lattice constants a and b reoptimized in each case. The results reveal that introducing non-uniform inclination leads to a locally stable structure with an energy minimum around $\theta'_{\text{incl}} \approx 1.6^\circ$ and $\varphi'_{\text{incl}} \approx 252^\circ$ (marked by a black circle). While increasing the degrees of freedom generally should reduce total interaction energy, the effect here is relatively small, indicating the robustness of the original model.

Next, we optimized $E_{\text{int(near)}}$ to examine the effect of interlayer stacking on the stability and competition between Type-II and Type-IV structures in tetracene. We treated θ'_{incl} as a variable for both structures while fixing φ'_{incl} at the minimum identified in Fig. 8b. The results, shown in Fig. 8c, indicate no significant stabilization of the Type-II structure due to inclination. In contrast, the Type-IV structure exhibits notable stabilization, with a minimum at $\theta'_{\text{incl}} \approx 2.5^\circ$, slightly larger than the optimized monolayer value based only on intralayer interactions. This finding demonstrates that interlayer stacking, combined with non-uniform long-axis inclinations, enhances the stability of Type-IV structures, whereas Type-II structure remains unaffected by this modification.

Figure 8d summarizes the calculated $E_{\text{int(near)}}$ values for Type-II and Type-IV structures in tetracene, pentacene, and hexacene, both before and after introducing non-uniform long-axis inclinations. Stabilization due to inclination is more pronounced in shorter polyacene of tetracene, where the Type-IV structure becomes more stable than Type-II. However, in longer polyacenes such as pentacene and hexacene, Type II remains the more stable form. These results align well with experimentally observed structures of tetracene and the high-temperature phase of pentacene.

The observed ring-number dependence provides valuable insights into the packing behavior of polyacenes: Shorter polyacenes tend to stabilize through deviations from parallel molecular alignment, while longer polyacenes favor a near-parallel configuration. Based on these findings, we conclude that the stability of Type-IV structures primarily arises from non-uniform long-axis inclinations introduced through interlayer stacking—a feature less significant in Type-II structures.

2.6. Comparison with Experimental Structures Considering Distant Intermolecular Interactions.

In this section, we compare the calculated structures obtained through stepwise optimization with experimentally observed crystal structures of polyacenes. First, we examine polymorph emergence by comparing the relative energies of different crystal phases derived from our calculations. As we described in Section 2.2, the interaction energy of monomolecular layers ($E_{\text{intra(8)}}$) follows the order: $R < N < G$ for all the polyacenes (Table 1). This energy hierarchy remains unchanged in pentacene and hexacene even when interlayer interactions with adjacent molecules are included. However, the lattice volume of the Type-I structure, derived from the *R*-form arrangement, is significantly larger than that of other polymorphs (see Table 1). This implies that dispersion interaction with distant molecules may not be negligible when comparing lattice energies.

To obtain a more accurate estimate of dispersion stabilization, we extended our calculations to include interactions with more distant molecules. To this end, we redefine the intermolecular interactions with the first-nearest group of molecules as

$$E_{\text{int(G1)}} = E_{\text{intra(8)}} + 2 \times E_{\text{inter(9)}},$$

and then calculated intermolecular interactions with more distant molecular groups from the second-nearest group ($E_{\text{int(G2)}}$) to the fifth-nearest group ($E_{\text{int(G5)}}$). Molecules in the first-nearest group, the second-nearest group, and the third-nearest group are illustrated in Fig. S8. Finally, we extrapolated convergence behavior to estimate the total intermolecular interactions across the entire crystal, denoted as:

$$E_{\text{int(all)}} = E_{\text{int(G1)}} + E_{\text{int(G2)}} + E_{\text{int(G3)}} + E_{\text{int(G4)}} + E_{\text{int(G5)}} + \dots,$$

as shown in Fig. S9. The results, summarized in Table 1, revealed that when long-range

interactions are considered, the $E_{\text{int(all)}}$ values of Type-II, III, and IV structures can be lower (more stable) than Type-I. Furthermore, the most stable structure is Type III in naphthalene and anthracene, Type IV in tetracene, and Type II in pentacene and hexacene, which are quite consistent with the actual polymorph tendencies. This finding suggests that the emergence of different polyacene polymorphs in actual crystals can be well explained by the present calculations. In particular, these results highlight the importance of considering long-range interactions to accurately assess the relative stability of different crystalline phases. While nearest-neighbor interactions primarily determine locally stable molecular arrangements, distant interactions influence overall phase stability.

Next, we evaluated the alignment between optimized and experimental structures. Tables S1 and S2 compare the structural parameters derived from our optimized structures with experimental crystal structures. The results show good overall agreement, which confirms the accuracy of our optimization approach. We also calculated powder diffraction patterns^{77,78} from our optimized model structures and compared them with experimental diffraction data, as presented in Fig. 9. For each structure, the diffraction angles and relative peak intensity ratios closely match experimental results, further validating the consistency between our optimized and experimental structures.

From the diffraction pattern comparisons, we find the following features: Type-III structures (naphthalene and anthracene) and Type-II structures (pentacene and hexacene) exhibit good agreement with experimental data. In contrast, Type-IV structures (tetracene and pentacene) and Type-I structure (pentacene) show some discrepancies at higher diffraction angles. These discrepancies are likely due to small differences in in-plane lattice constants at finite temperatures. The deviation of the *R* form in experimental pentacene (Type-I structure) will be discussed in the next section.

2.7. Deviation of the *R*-Form in Actual Pentacene.

The optimized *R* form of pentacene is metastable and closely resembles the actual Type-I (thin-film phase) structure. However, a slight but clear deviation in molecular arrangement exists between the *R*-form and the reported crystal structure. Specifically, the actual Type-I structure exhibits a subtle tilt in the molecular long axis, though this deviation is significantly smaller than in the *N* or *G* forms. More importantly, crystalline symmetry differs between the *R* form and the actual crystal structure, as illustrated in Fig. 10.

The actual Type-I structure, commonly known as the thin-film phase of pentacene, can be achieved by introducing a slight inclination of the molecular long axis while maintaining space inversion symmetry. Note that the pentacene thin-film phase exists as uniaxially oriented polycrystalline thin films on substrates. Structural analysis was performed on these films, though

various c -axis lengths have been reported depending on film fabrication conditions.^{79–82}

To explore the stability of the Type-I structure, we performed structural optimizations using $(\varphi'_{\text{incl}}, \theta'_{\text{incl}})$ to independently adjust molecular positions within the unit cell. Our results showed that $E_{\text{intra}(6)}$ value consistently increases, indicating that the layer becomes less stable when these variables are introduced (Fig. 10d). In contrast, $E_{\text{inter}(7)}$ decreases slightly, suggesting that interlayer interactions provide minor stabilization through molecular long-axis shifts (Fig. 10e). However, $E_{\text{int(near)}}$ consistently increases across all displacement configurations, indicating no local energy minimum near the Type-I structure. These findings indicate that the metastability of the R -form arise from its sensitivity to interactions with substrate surfaces.^{83–87}

2.8. Calculation of Transfer Integrals.

Intermolecular transfer integrals, calculated from the overlap of highest occupied molecular orbitals (HOMOs), are essential for understanding charge carrier transport in OSCs. We computed the transfer integrals at T -shaped and slipped-parallel contacts for each optimized structure, ensuring close alignment with experimentally observed structures.

For pentacene, the Type-I structure exhibited the largest transfer integrals, reflecting excellent two-dimensional charge transport (Fig. 11a). In contrast, transfer integrals were lower in Type-II and Type-IV structures compared to those in Type-I. Our analysis reveals that transfer integrals at T -shaped contacts are largest when no long-axis shift occurs but decrease significantly when long-axis shift is introduced. This effect is more pronounced in Type-IV structures than in Type-II, with similar trends observed across polyacenes of varying ring numbers.

To further investigate the origins of these trends, we calculated transfer integrals for metastable structures during the optimization process (Sections 2.1 and 2.2). Specifically, we examined the dependence of transfer integrals on the dihedral angle (α) (Fig. 11b) and the inclination angle (θ_{incl}) (Fig. 11c). The results show that transfer integrals are highest without long-axis misalignment but change drastically with increasing θ_{incl} . The integrals diminish when the misalignment corresponds to half a benzene ring but recover when the misalignment reaches one benzene ring. This behavior is attributed to the overlap of molecular orbital antinodes, which maximizes transfer integrals. These findings provide a consistent explanation for the differences in transfer integrals between Type-I, Type-II, and Type-IV structures. Specifically, Type-II structures exhibit relatively large transfer integrals because θ_{incl} (25.4° in hexacene) is close to the optimal misalignment of one benzene ring. These variations in transfer integrals highlight the importance of careful design of molecules and crystals in maximizing charge transport.

III. DISCUSSION

The crystal structures of molecular solids are fundamentally determined by configuration

parameters that minimize the total intermolecular interaction energies within the crystal.^{11–15} However, identifying the optimal combination of these parameters is generally challenging due to the vastness of the configurational space.^{40–50} In this study, we focused on the layered crystalline nature of a series of rigid, rodlike polyacenes, which allowed us to employ a structural model with well-defined configuration parameters. We began with an arbitrarily arranged two-dimensional molecular layer under glide symmetry, where all molecules were aligned parallel and unshifted along their long axes. We then introduced additional degrees of freedom to account for molecular tilting, torsion, and interlayer stacking—ultimately defining 11 parameters in total (α , a , b , φ_{incl} , θ_{incl} , θ_{twist} , φ'_{incl} , θ'_{incl} , x , y , and z).

The stability and formation of each polymorph were investigated using a combination of vdW contact models and DFT-D calculations. Our results demonstrate that a stepwise optimization approach is both effective and practical: the intralayer arrangement is determined first, followed by the optimization of interlayer configurations. When interlayer interactions are relatively strong, re-optimizing the intralayer geometry while incorporating interlayer effects further improves the reproduction of experimental structures. Additionally, considering interactions with more distant molecules enhance the accuracy of relative polymorph stability. Using this method, we successfully reproduced all four experimentally observed polymorphs of polyacenes, capturing both their structural details and their stability trends as a function of ring number. Potential map analyses visualizing changes in the sum of interaction energies (e.g., $E_{\text{intra}(6)}$ or $E_{\text{int}(8)}$) during the optimization process further clarified the mechanisms stabilizing each polymorph. Key findings and challenges encountered throughout the optimization are discussed below.

Role of Glide Symmetry. Crystal symmetry plays a critical role in simplifying configurational complexity. Glide symmetry, one of the most prevalent symmetries in organic molecular crystals,⁸⁸ allows for the unification of several parameters that would otherwise introduce independent molecular displacements. This reduction in degrees of freedom also contributes to stabilization, especially via entropy gain at finite temperatures. Indeed, lower molecular weight polyacenes such as naphthalene and anthracene tend to exhibit higher symmetry, highlighting the importance of thermal motion. From this perspective, it is reasonable to assume high symmetry in the initial optimization step, particularly given that all calculations were performed on static structures at 0 K. Notably, assuming glide symmetry enables the construction of a versatile molecular pair that serves as the fundamental unit for building the entire crystal structure through simple translational operations, as demonstrated in this study.

Selection of Configuration Parameters. The choice of configuration parameters is crucial in the stepwise optimization framework. The two-dimensional molecular arrangement used in the initial step is highly versatile, enabling the exploration of stable structures over a broad configurational space and naturally leading to the *R*-form structure. The subsequent uniform tilting of the

molecular long axes generates the *G*- and *N*-form arrangements, which serve as essential starting points for describing polyacene crystal structures.

Torsional deformation and non-uniform tilting of the molecular long axes were introduced based on comparisons with experimental crystal structures of polyacenes, although these parameters also maintain general applicability. Notably, the inclusion of these distortions enabled successful optimization of the Type-III and Type-IV structures, yielding good agreement with experimentally observed crystal data. These parameters were crucial for identifying the origins of each crystal structure and understanding their stabilization mechanisms.

However, for molecules whose crystal structures are unknown, a full exploration of the entire configuration space would be necessary—presenting a significant challenge for predictive applications. To overcome this limitation, it will be essential to apply the current approach to a broader range of systems and accumulate empirical insights.

Stability and Deviation of the Type-I Crystal. The *R*-form arrangement produces the most stable and symmetric monomolecular layer. However, its interlayer interactions are weaker compared to the *G*- and *N*-form structures. When long-range interactions are taken into account, these polymorphs become energetically more favorable than the *R* form. Indeed, the Type-I structure has not been confirmed in single crystals and has only been observed in thin films on substrates.^{23,76} This suggests that genuine *R*-form crystals may transiently form as ultrathin crystal flakes with a few molecular layers thickness in solution or gas-phase environments but are easily transform into thicker bulk-phase crystals. On the other hand, ultrathin films are necessarily influenced by the substrate, where the actual Type-I structure reflect substrate-induced effects or front-back asymmetry at the interface, resulting in deviations from the ideal *R*-form.^{79–87} In such cases, the crystal structure becomes highly sensitive to crystal quality, complicating accurate structure determination. Actually, previous studies have indeed reported correlations between pentacene packing and substrate conditions.

Polyacene crystals represent ideal systems for the application of the stepwise optimization method, owing to their pronounced layered crystallinity and relatively high symmetry. Based on our findings, we summarize below the key structural characteristics of each polymorph identified in the polyacene series:

Type-I structure: A single-layer structure corresponding to the *R*-form herringbone packing. Although it is the most stable monolayer configuration, weak interlayer interactions reduce its relative stability when long-range interactions are considered. The experimentally observed structure exhibits lower symmetry and slight deviations from the calculated ideal, due to substrate or interfacial effects.

Type-II structure: An *N*-form layered herringbone structure featuring partially offset *T*-shaped

contacts (by one ring). Rodlike molecules form close-packed, parallel arrays with relatively strong interlayer interactions. This packing is energetically favored in polyacenes with larger ring numbers ($n = 5, 6$), consistent with experimental observations.

Type-III structure: A *G*-form layered structure in which *T*-shaped contacts are uniformly shifted by one ring. Molecules exhibit torsional deformation, with their long axes tilted antiparallel with preserving projected symmetry. This enhances interlayer interactions and results in the most stable packing for small ring polyacenes ($n = 2, 3$), as seen experimentally.

Type-IV structure: An *N*-form layered structure distinct from Type II, featuring a stacking arrangement where the long-axes at *T*-shaped contacts shift from fully parallel to slightly offset. This leads to increased interlayer interactions and stabilizes the structure. Type IV is the most stable polymorph for tetracene ($n = 4$), which has an intermediate ring number.

METHOD

The intermolecular interaction energy was evaluated by DFT-D calculation. Gaussian16 program⁸⁹ was used for DFT calculations. The B3LYP functional,⁹⁰ 6-311G** basis set and Grimme's D3 dispersion correction⁶⁹ were used for the DFT-D calculations. The optimized geometries of isolated molecules by the DFT-D calculations were used for the calculations of intermolecular interaction energies. Basis sets superposition error (BSSE) was corrected by a counterpoise method.⁹¹ An energy decomposition analysis was carried out using SAPT(DFT) with the PBE0-GRAC.^{70,71,92} Psi4 program was used for SAPT(DFT) calculations.^{93,94} The aug-cc-pVDZ basis set was used for the SAPT(DFT) calculations. Total interaction energy by SAPT(DFT) (E_{sapt}) was calculated as the sum of the calculated each term by the SAPT(DFT). The transfer integrals between HOMOs were calculated using the molecular orbitals obtained by Gaussian16 program at the B3LYP/6-31G* level.

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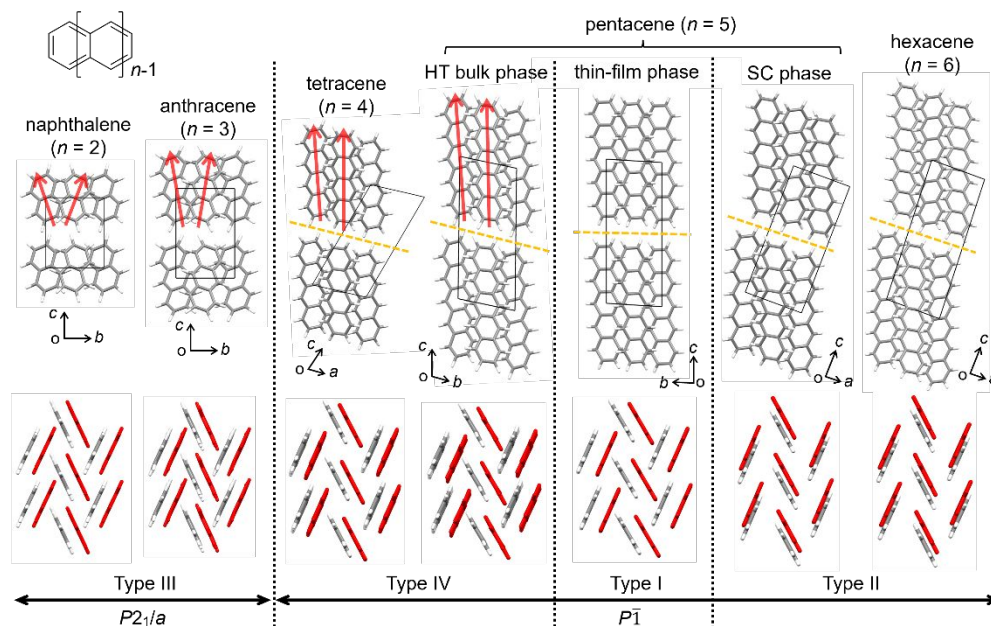


Fig. 1. Crystal structures of polyacenes and their classification into four structural types. This figure illustrates the crystal structures of naphthalene,²⁰ anthracene,²¹ tetracene,¹⁸ hexacene,¹⁹ and three polymorphs of pentacene (high temperature bulk phase,²² thin-film phase,¹⁷ and single-crystal phase¹⁸). These structures are categorized into four distinct types (Type I, II, III, and IV), based on crystal symmetry, the inclination of molecular long axes within the layers, and the interlayer stacking arrangements. The upper panels show the intralayer molecular arrangements, viewed along an axis parallel to the layer interface and perpendicular to the slipped-parallel contacts within the layers. The red arrows indicate the direction of molecular long axis. The lower panels depict the interlayer stacking configurations, with the view aligned parallel to the molecular planes. Molecules shown by gray and red color show those in the overlayer and the underlayer, respectively. This classification highlights the diversity in polyacene packing motifs depending on the number of fused rings.

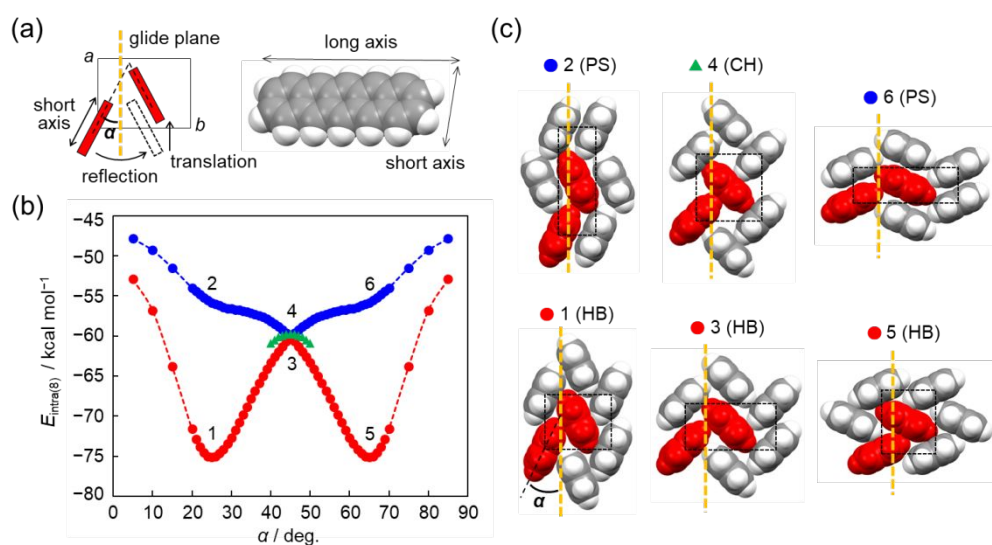


Fig. 2. Optimization of regular intralayer molecular arrangements in polyacenes. (a) Schematic representation of a basic unit consisting of a molecular pair, with the molecular short axis indicated by a rectangle. The arrangement is characterized by three key parameters: the angle α between the molecular plane and the glide plane and the lattice constants a and b . (b) Sum of intermolecular interaction energies of a molecule with eight nearest neighbors ($E_{\text{intra}(8)}$), calculated for stable and metastable molecular arrangements of pentacene. a and b are optimized for each α . The obtained arrangements are categorized by the packing features into herringbone (HB), π -stacked (PS), and crosshatch (CH) packings, plotted by red, blue, and green colors, respectively. (c) Visualizations of the molecular arrangements corresponding to labels 1–6 in (b), viewed along the molecular long axes. Arrangements 1, 3, and 5 represent HB packing, 2 and 6 correspond to PS packing, and 4 depicts a CH structure.

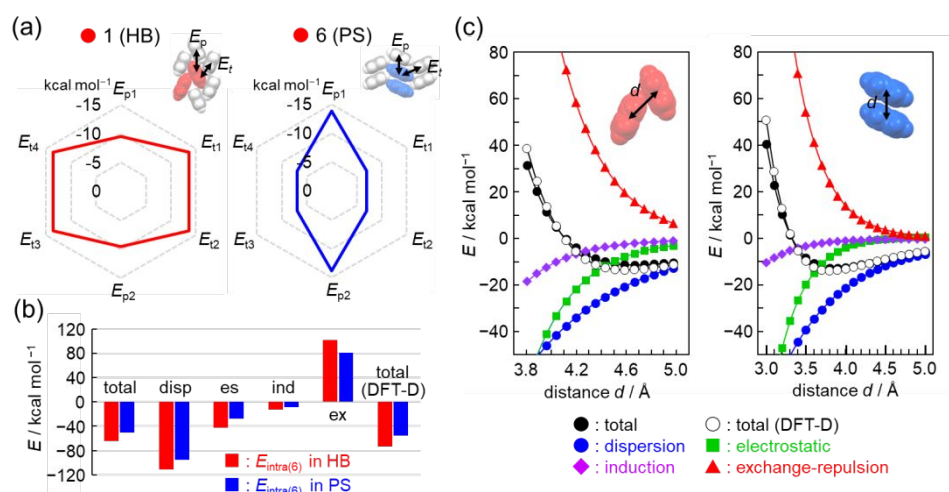


Fig. 3. Key characteristics of regular HB and PS arrangements in pentacene.

(a) Radar charts comparing the calculated intermolecular interaction energies at the T -shaped and slipped-parallel contacts in both the HB and PS arrangements at $\alpha = 25^\circ$. (b) Breakdown of the contributions from electrostatic, induction, dispersion, and exchange-repulsion energies to the total interaction energies with the six nearest-neighbors ($E_{\text{intra}(6)}$) for both HB and PS arrangements at $\alpha = 25^\circ$. (c) Dependence of each interaction energy on intermolecular distance, comparing the T -shaped contact (E_t) in the HB arrangement and the slipped-parallel contact (E_p) in the PS arrangement at $\alpha = 25^\circ$.

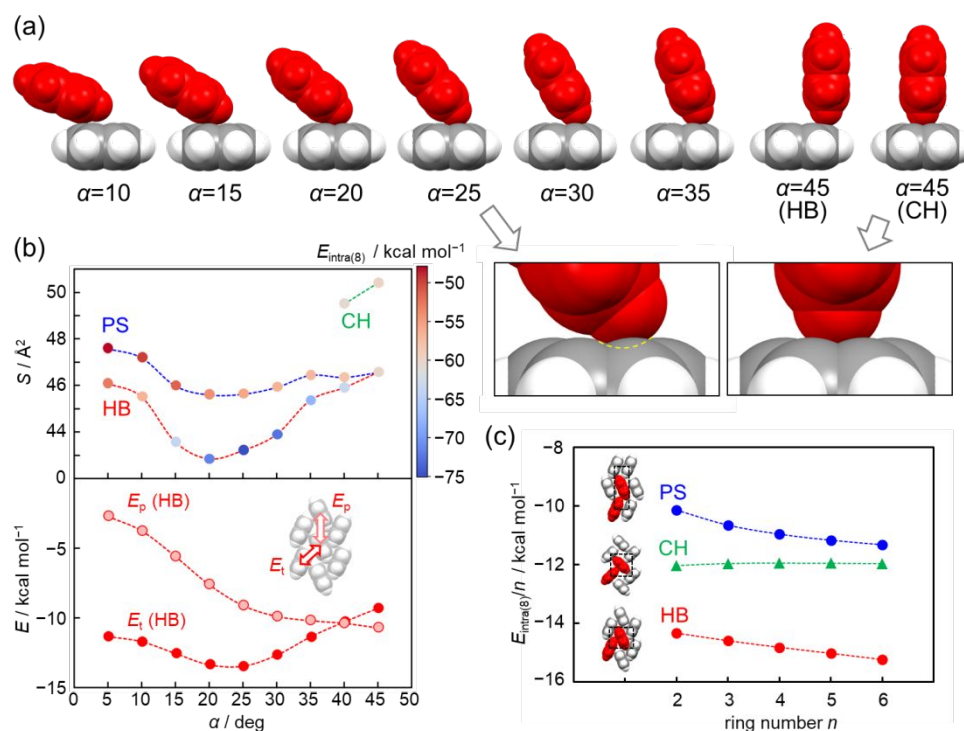


Fig. 4. Origins of the stability of regular HB packing in polyacenes. (a) Evolution of T-shaped contacts in the metastable HB packing as a function of α and in CH packing at $\alpha = 45^\circ$. Expanded images are shown for $\alpha = 25^\circ$ in HB and $\alpha = 45^\circ$ in CH in the right lower panel. (b) Top panel: Dependence of unit cell area ($S = a \times b$) on α for the HB, PS, and CH packing layers. Bottom panel: Intermolecular interaction energies at T-shaped (E_t) and slipped-parallel (E_p) contacts in the HB packing as a function of α . (c) Ring-number dependence of $E_{\text{intra}(8)}/n$ for the HB ($\alpha = 25^\circ$), PS ($\alpha = 25^\circ$), and CH ($\alpha = 45^\circ$) packings.

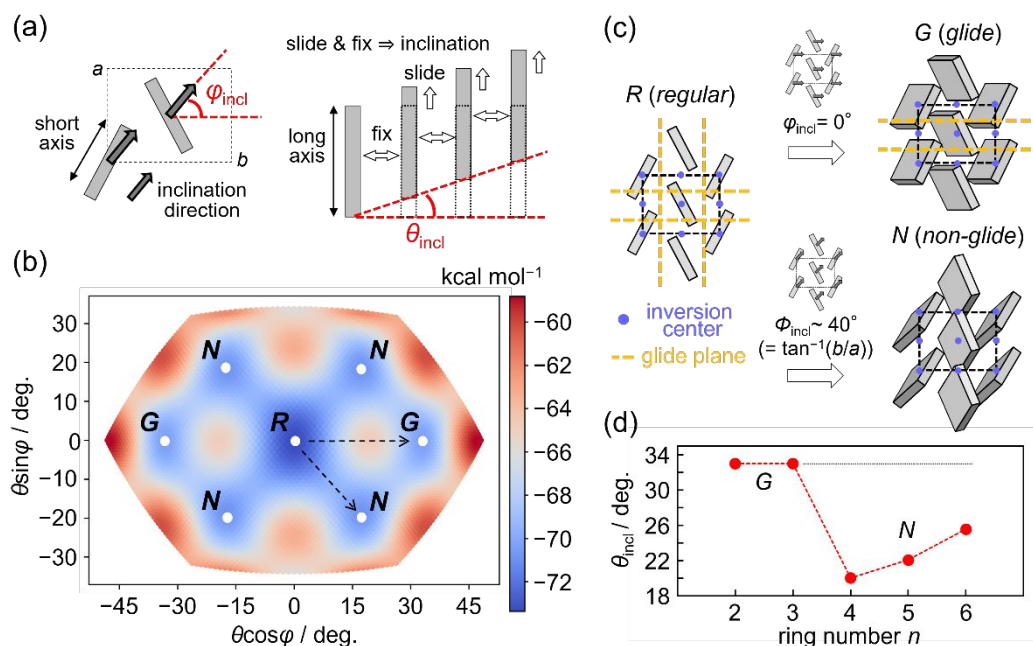


Fig. 5. Structure optimization through uniform long-axis inclination. (a) Schematic representation of uniform long-axis inclination, where the molecular layer maintains a flat surface and constant interplanar spacing between adjacent molecules. The inclination direction within the layer plane is denoted by ϕ_{incl} , while θ_{incl} represents the angle of inclination relative to the normal of the layer surface. (b) Calculated $E_{\text{intra}(6)}$ landscape for the pentacene monomolecular layer with varying ϕ_{incl} and θ_{incl} , showing the potential minima corresponding to the *R*, *G*, and *N* forms. $E_{\text{intra}(6)}$ is the sum of the interaction energy of a molecule with six nearest-neighbor molecules in the same layer. (c) Schematics illustrating the molecular packing features of the *R*, *G*, and *N* forms, highlighting the distinct inclination characteristics of each packing type. (d) Ring-number dependence of θ_{incl} in the *G* and *N* forms, demonstrating how the inclination angle evolves with increasing molecular size in polyacenes.

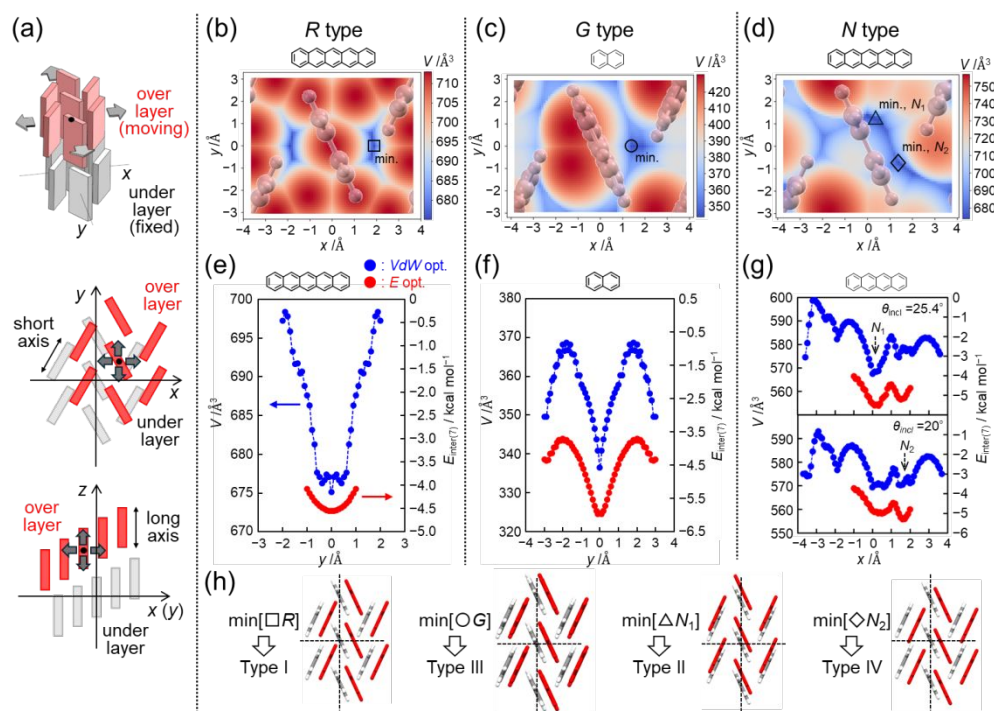


Fig. 6. Optimization of interlayer stacking arrangements. (a) Schematic for searching optimized lateral coordinates in the interlayer stacking arrangements of monomolecular layers. The origin of the coordinate is set where the overlayer and the underlayer are identical with respect to the parallel translation along the molecular long axis. (b–d) Upper panels show calculated unit cell volume at the vdW contacts between the monomolecular layers of (b) the *R*-form pentacene, (c) the *G*-form naphthalene, and (d) the *N*-form pentacene, plotted as functions of relative in-plane coordinates along the layer surface. Lower panels show calculated interlayer interaction energy ($E_{\text{inter}(7)}$, red dots) and unit cell volume (blue dot), plotted as functions of y (or x) shift along the local minimum valleys in the respective volume maps shown in the upper panels. In the lower panel of (d), calculations are done for the *N*-form tetracene (at $\theta_{\text{incl}} = 25.4^\circ$ and 20.0°). (e) The relative interlayer arrangement shown between the overlayer (red) and the underlayer (gray), viewed along the molecular long axis.

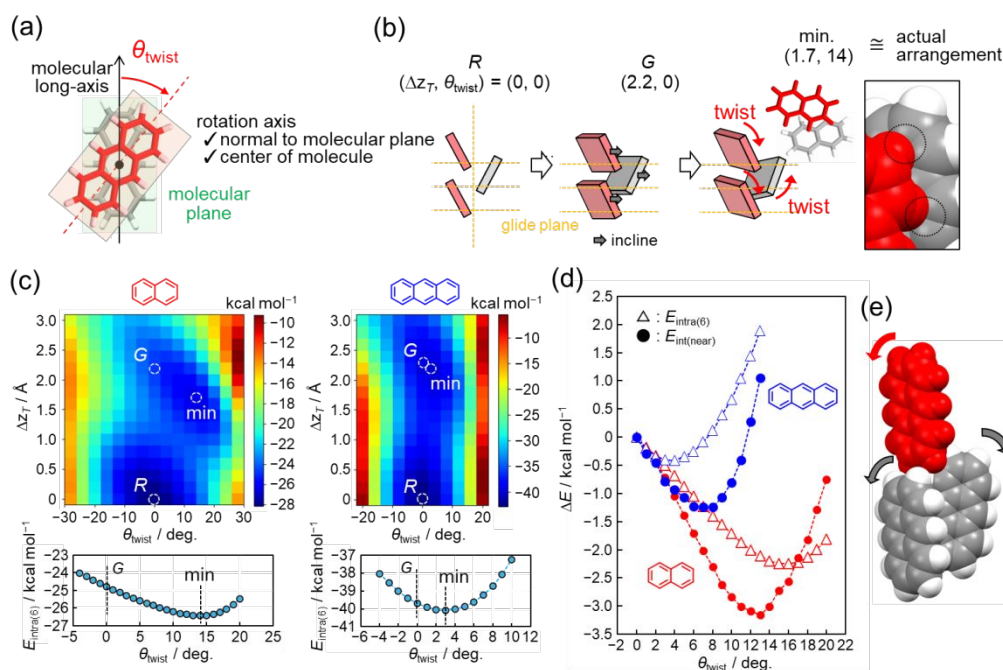


Fig. 7. Optimization of long-axis twist shift under glide symmetry. (a) Schematic for rotation of molecular long axis around the axis perpendicular to the molecular plane, denoted as θ_{twist} . (b) Schematic for a twist shift, where the long-axis inclination parallel to the glide plane is coupled with twist rotation of adjacent molecules at the T-shaped contacts in the opposite direction. The glide symmetry is preserved by the operation. Right: view of intermolecular T-shaped contact in the finally optimized arrangement of naphthalene by the twist operations, which eventually enhances another CH/ π contact between benzene ring and hydrogen atom. (c) Variation of calculated interaction energy ($E_{\text{intra}(6)}$), plotted as functions of θ_{twist} and Δz_T , for naphthalene (left panel) and anthracene (right panel), with the cross section plotted along the local minimum valleys in the lower panel. (d) Variations of $E_{\text{intra}(6)}$ (open triangles) and $E_{\text{int(near)}}$ (filled circles) of naphthalene (red) and anthracene (blue). In the calculation of $E_{\text{int(near)}}$, interlayer distance is optimized at each point by minimizing the interaction energy. Relative interlayer arrangements between the overlayer molecule (red) and the underlayer molecules (gray) are shown in the right.

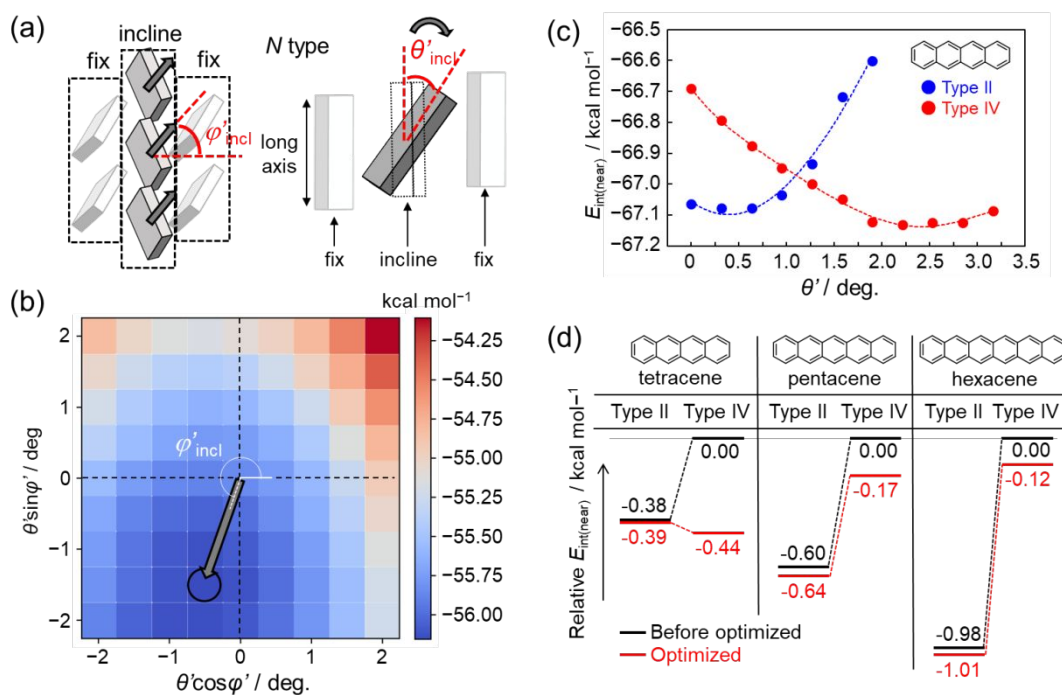


Fig. 8. Optimization with additional non-uniform long-axis inclination. (a) Schematic for independent long-axis inclination of molecule in the unit cell. ϕ'_{incl} and θ'_{incl} are the inclination direction within the layer plane and the inclined angle from the normal to the layer surface, respectively. (b) Variation of calculated interaction energy ($E_{intra(6)}$) at arbitrary independent long-axis inclination with ϕ'_{incl} and θ'_{incl} for tetracene monomolecular layer. (c) Variation of calculated interaction energy ($E_{int(near)}$) by independent long-axis inclination for the interlayer stacking at N_1 (Type II, blue filled circle) and at N_2 (Type IV, red filled circle). (d) Variation of interaction energy ($E_{int(near)}$) for Type II and Type IV stacking in tetracene, pentacene, and hexacene, with and without the optimization by additional independent long-axis inclination.

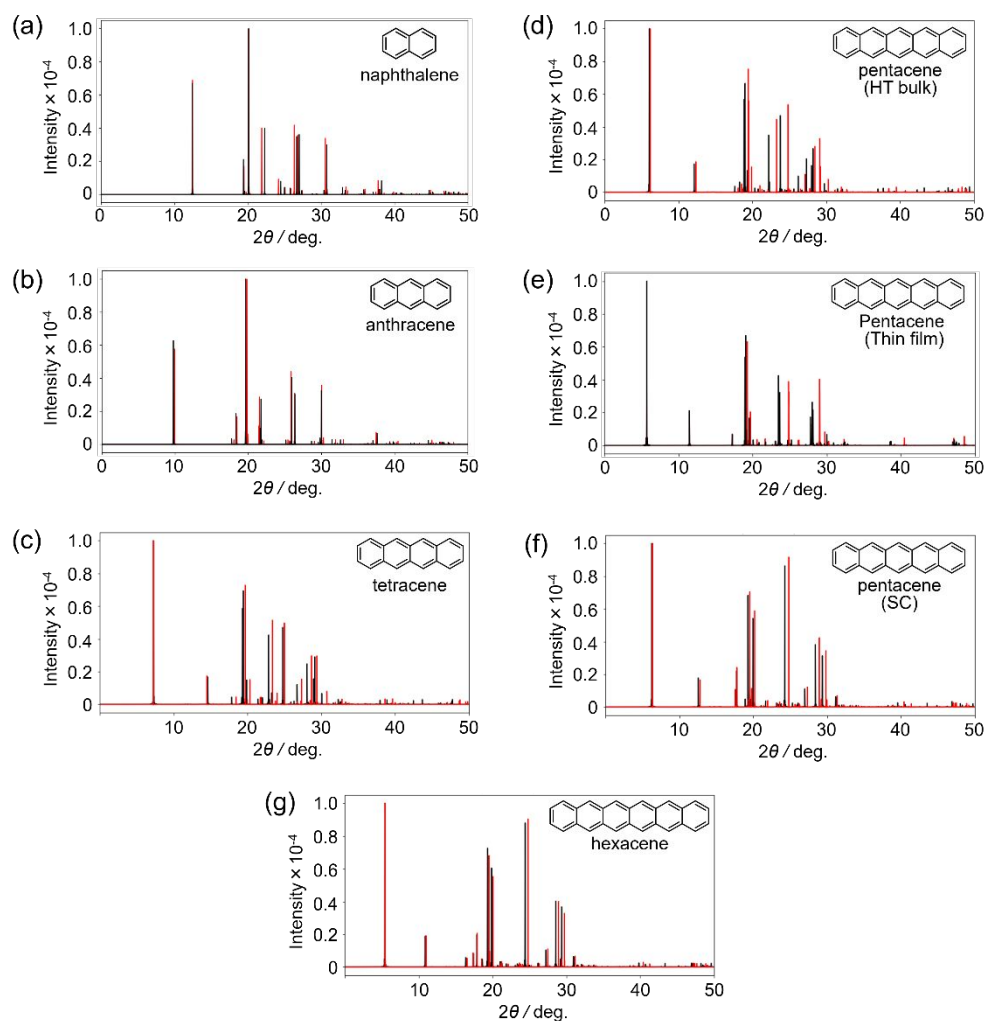


Fig. 9. Comparisons of x-ray powder diffraction patterns between actual (black) and calculated (red) crystal lattices. (a) naphthalene, (b) anthracene, (c) tetracene, (d) pentacene (high temperature bulk phase), (e) pentacene (thin film phase), (f) pentacene (single crystal phase), and (g) hexacene.

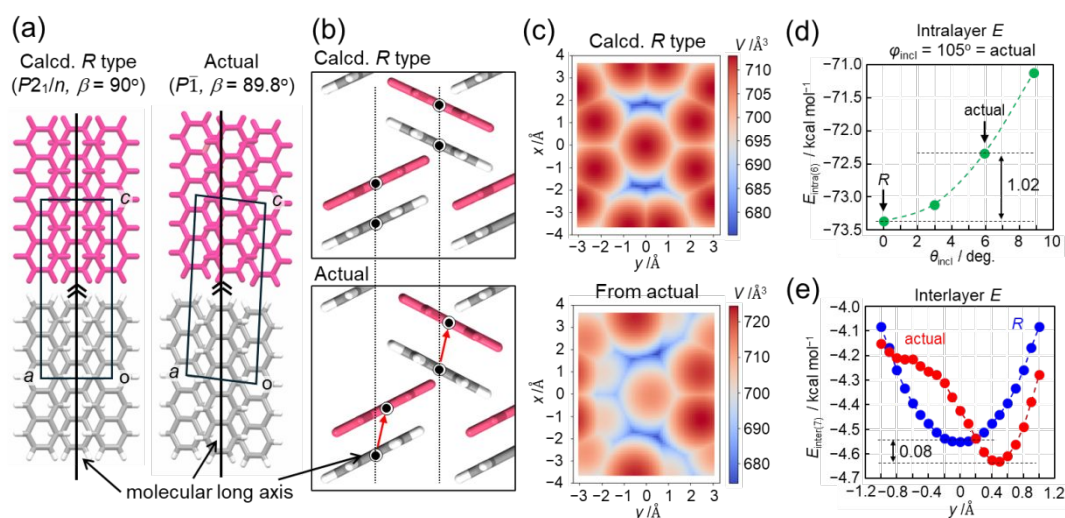


Fig. 10. Difference between calculated *R*-form and actual crystal lattices of pentacene. (a) View along *b* axis, and (b) view along molecular long axis. (c) Calculated unit cell volume at the vdW contacts. Slight difference is observed in both the intralayer molecular arrangements and the stable interlayer stacking arrangements. (d) Calculated intralayer interaction energy ($E_{\text{intra}(6)}$), plotted as a function of θ_{incl} at $\varphi_{\text{incl}} = 105^\circ$. (e) Calculated interlayer interaction energy ($E_{\text{inter}(7)}$), plotted as a function of y shift along the minimum valley in (c).

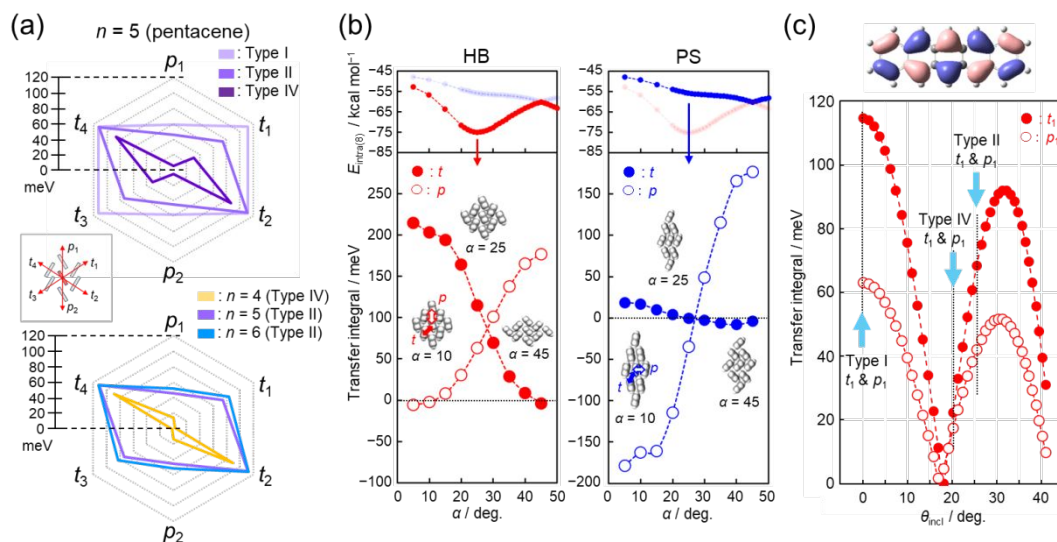


Fig. 11. Calculated intermolecular transfer integrals of optimized crystal lattices and comparison between polymorphs. (a) Upper panel: Comparison of intermolecular transfer integrals in Type I, II, and IV polymorphs of pentacene. Lower panel: Comparison of intermolecular transfer integrals in tetracene (Type IV), pentacene (Type II), and hexacene (Type II). (b) α dependence of intermolecular transfer integrals at the T -shaped (t) and slipped parallel (p) contacts for stable and metastable HB (left) and PS (right) packings of pentacene monomolecular layer. Intralayer interaction energy ($E_{\text{intra}(8)}$) of these packings (Fig. 2b) are shown in the upper panels. (c) Calculated intermolecular transfer integrals at the T -shaped and slipped parallel contacts for pentacene monomolecular layer, plotted as a function of θ_{incl} . The θ_{incl} values in the respective polymorphs are marked in the plot. The transfer integrals crucially depend on the intermolecular overlap between HOMOs, depicted in the upper panel.

Molecule	Crystal type	Sum of Intermolecular Interaction Energy E (kcal/mol)						Near Interaction Ratio	Volume ($\text{\AA}^3$)
		Intralayer $E_{\text{intra}(8)}$	Interlayer $2 \times E_{\text{inter}(9)}$	Total (near) $E_{\text{int}(G1)}$	Intralayer Interaction Ratio $E_{\text{intra}(8)} / E_{\text{int}(G1)}$	Total $E_{\text{int}(\text{all})}$			
Naphthalene ($n = 2$)	Type III (act ^d)	-27.25	-12.95	-40.20	0.678	-42.73	0.941	335.70	
	Type I	-28.67	-9.58	-38.25	0.750	-40.41	0.947	349.92	
Anthracene ($n = 3$)	Type III (act)	-40.18	-13.72	-53.90	0.745	-57.28	0.941	446.93	
	Type I	-43.78	-9.81	-53.59	0.817	-56.70	0.945	453.60	
Tetracene ($n = 4$)	Type IV (act)	-57.14	-12.09	-69.23	0.825	-73.80	0.938	547.44	
	Type II	-56.91	-11.92	-68.83	0.827	-73.56	0.936	552.35	
Pentacene ($n = 5$)	Type I	-59.26	-9.86	-69.12	0.857	-73.17	0.944	561.60	
	Type II (act)	-72.73	-12.11	-84.84	0.857	-90.66	0.936	656.26	
	Type I (act ^b)	-75.08	-9.90	-84.98	0.884	-90.03	0.944	665.28	
	Type IV (act)	-72.65	-11.99	-84.64	0.858	-89.96	0.941	662.25	
Hexacene ($n = 6$)	Type II (act)	-88.98	-12.09	-101.07	0.880	-107.91	0.937	762.46	
	Type I	-91.35	-9.84	-101.19	0.903	-107.39	0.942	762.54	
	Type IV	-88.43	-11.96	-100.39	0.881	-106.77	0.940	770.34	

Table 1. Summary of the calculated total intermolecular interaction energies and unit cell volumes for various polymorph types of all polyacenes, obtained via stepwise optimization calculations. Values are given for the intralayer ($E_{\text{intra}(8)}$) and interlayer ($2 \times E_{\text{inter}(9)}$) components, as well as their sum ($E_{\text{int(G1)}} = E_{\text{intra}(8)} + 2 \times E_{\text{inter}(9)}$) along with the ratios of intralayer interaction ($E_{\text{intra}(8)}/E_{\text{int(G1)}}$), where the calculations consider interactions with the first-nearest group of molecules. Then values are given for the sum of intermolecular interactions including more distant molecules throughout the entire crystal ($E_{\text{int(all)}}$), along with the ratios of interactions with first-nearest group of molecules ($E_{\text{int(G1)}}/E_{\text{int(all)}}$). $E_{\text{int(all)}}$ were estimated using exponential functions of the form $A + B \times \exp(-CX)$, where A , B , and C are the fitting parameters, while X is the classification index representing the distance between the central molecule and its surrounding molecules as depicted in Fig. S8. The intermolecular interactions corresponding to $X = 1, 2, 3, 4$, and 5 for each molecule and crystal type were calculated to determine the fitting parameters A , B and C .

a. “act” indicates that this polymorph type occurs in actual crystals.

b. The crystal symmetry differs from that of the actual crystal.